Lecture 12: Meteorites, Asteroids, Minor Planets & Comets

Planetary Systems

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

2026

Comet 1P/Halley, Giotto Halley Multicolour Camera (1986). Credit: ESA

Learning Objectives

By the end of this lecture, you will be able to:

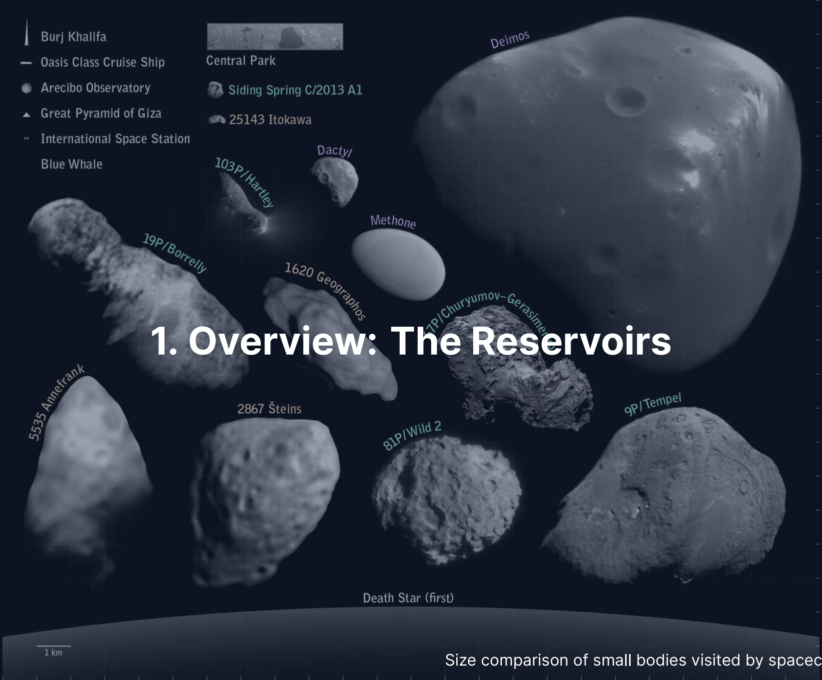
- Classify meteorites and use isotopic chronometers to date the early solar system.
- Describe the asteroid-belt and trans-Neptunian populations and their dynamics.
- Explain the origin and structure of comets.
- Derive the Pb-Pb isochron age of CAIs.
- Use small bodies as formation fossils that record dynamical and chemical history.

Everything that did *not* become a planet preserves more about how the solar system formed than the eight planets do.

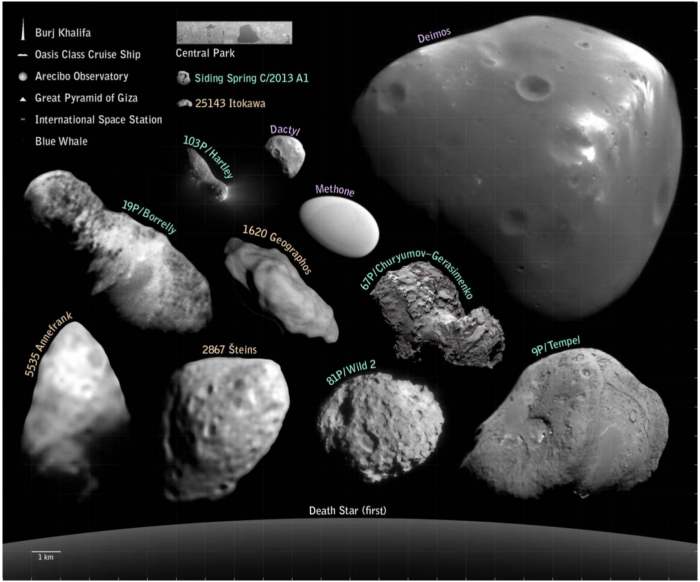
Recap: Where We Are in the Course

- L9-L11 walked through every major planet: rocky and giant.
- Today: the leftovers. Asteroids and meteorites inside Neptune; KBOs, dwarf planets, comets, Oort cloud beyond.
- Pluto is treated here as the largest known KBO, not as a planet.
- L13: exoplanets. L14: synthesis and astrobiology.

1. Overview: The Reservoirs



Size comparison of small bodies visited by spacecraft. NASA/JPL-Caltech



Small bodies visited by spacecraft, scaled against terrestrial landmarks. Credit: NASA/JPL-Caltech, public domain

Reading the Small-Bodies Inventory

Three great reservoirs:

- **Main belt** (2.1–3.3 AU): ~ 10^6 bodies > 1 km. Total mass ~ $4 \times 10^{-4}\ M_\oplus$ (Ceres alone is 30%).
- **Trans-Neptunian region** (Kuiper Belt + scattered disk + detached): > 4,000 catalogued, many more incoming with Rubin.
- **Oort cloud** (~ 2,000–50,000 AU): ~ 10^{11}–10^{12} objects inferred. *Never directly observed.*

Plus Trojan swarms at Jupiter's L4/L5 and a growing population of *interstellar* visitors.

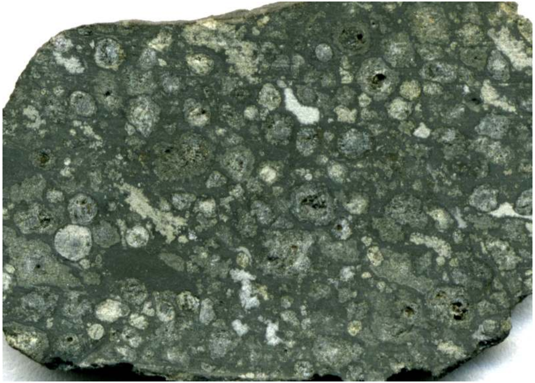
2. Meteorite Taxonomy

Allende meteorite (CV3 chondrite, fell 1969). J. St. John CC BY 2.0

Why Meteorites Matter

- ~70,000 catalogued specimens, mostly Antarctic + desert finds.
- **The oldest rocks accessible to a terrestrial laboratory.**
- Earth's record: oldest zircons ~ 4.0 Gyr; Moon dominated by impact melts.
- Most primitive meteorites contain disk-condensation grains, some *older than the Sun*.
- Anchor age: 4567.30 ± 0.16 Myr. **The most precise number in planetary science.**

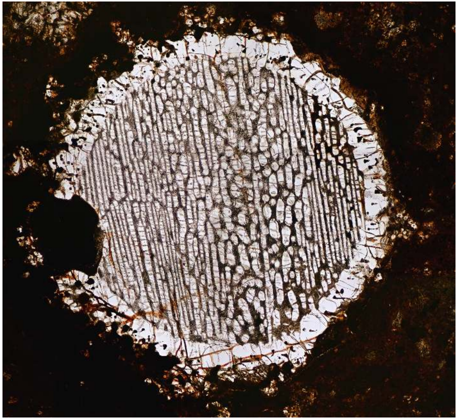
The First Cut: Chondrites vs. Everything Else



Allende slab: chondrules + bright CAIs in dark matrix

- **Chondrites**: never melted whole. Aggregates of disk solids.
- Characteristic: ~mm-scale silicate *chondrules*.
- Achondrites, irons, stony irons: from *differentiated* parent bodies.
- Differentiation needs heat → short-lived $^{26}\mathrm{Al}$.
- Iron meteorites = cores; achondrites = crusts/mantles.

Chondrules in Thin Section

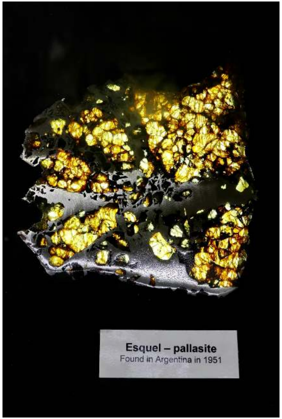


- Cross-polarised thin section, NWA 5930.
- Each chondrule: ~mm bead of olivine + pyroxene.
- Frozen from a fully molten droplet in seconds to hours.
- Peak $T \sim 1500$–1900 K.
- How chondrules form remains an unsolved problem.

Iron Meteorites: Reading a Cooling Rate

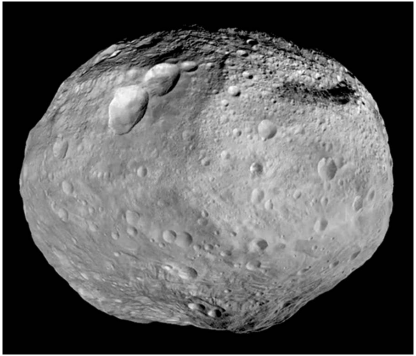
- Widmanstätten pattern: kamacite + taenite Fe-Ni lattice.
- Forms only at cooling rates $\lesssim 100\ \mathrm{K\ Myr^{-1}}$.
- Slow cooling possible only inside metallic cores of ~tens-of-km bodies.
- Lattice spacing → original parent-body radius.

Pallasite: A Core-Mantle Boundary in Hand



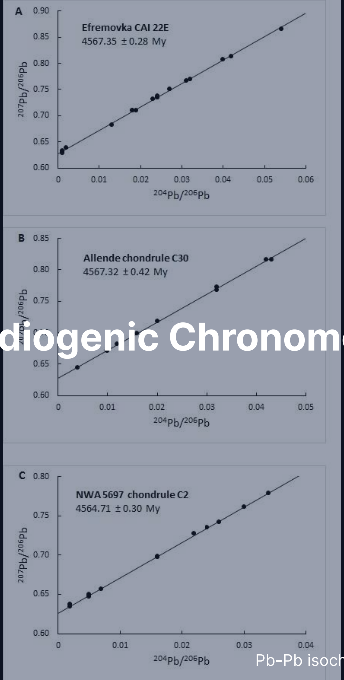
- Esquel pallasite: yellow-green olivine in silvery Fe-Ni metal.
- Texture suggests boundary between molten core and silicate mantle.
- Olivine crystals sank into metal during a giant impact.
- **A direct sample of a destroyed planetesimal's core-mantle boundary.**

Vesta: HED Parent Body



- HED meteorites (howardites/eucrites/diogenites) = ~ 6% of falls.
- Identical mineralogy + oxygen isotopes.
- Dawn (2011–2012): confirmed Vesta as parent.
- Rheasilvia basin (south pole): excavated and ejected the HEDs.
- Lunar + Martian meteorites identified the same way (oxygen isotopes + trapped Mars atmosphere).

3. Radiogenic Chronometers



Pb-Pb isochrons for CAI 22E and chondrules. Connelly et al. (2012)

The Decay-Clock Equation

Radioactive decay is the chronometer:

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N, \qquad N(t) = N_0\, e^{-\lambda t}$$

Half-life: $t_{1/2} = \ln 2/\lambda$.

- **Long-lived chronometers** ($t_{1/2} \gg$ age of solar system): $^{238}\mathrm{U}$, $^{235}\mathrm{U}$, $^{87}\mathrm{Rb}$, $^{147}\mathrm{Sm}$. Active today.
- **Short-lived (extinct)**: $^{26}\mathrm{Al}\,(0.72\,\mathrm{Myr})$, $^{53}\mathrm{Mn}\,(3.7)$, $^{182}\mathrm{Hf}\,(8.9)$, $^{129}\mathrm{I}\,(16)$, $^{244}\mathrm{Pu}\,(80)$.
- Long-lived → absolute age. Short-lived → sharp *relative* age within their decay window.

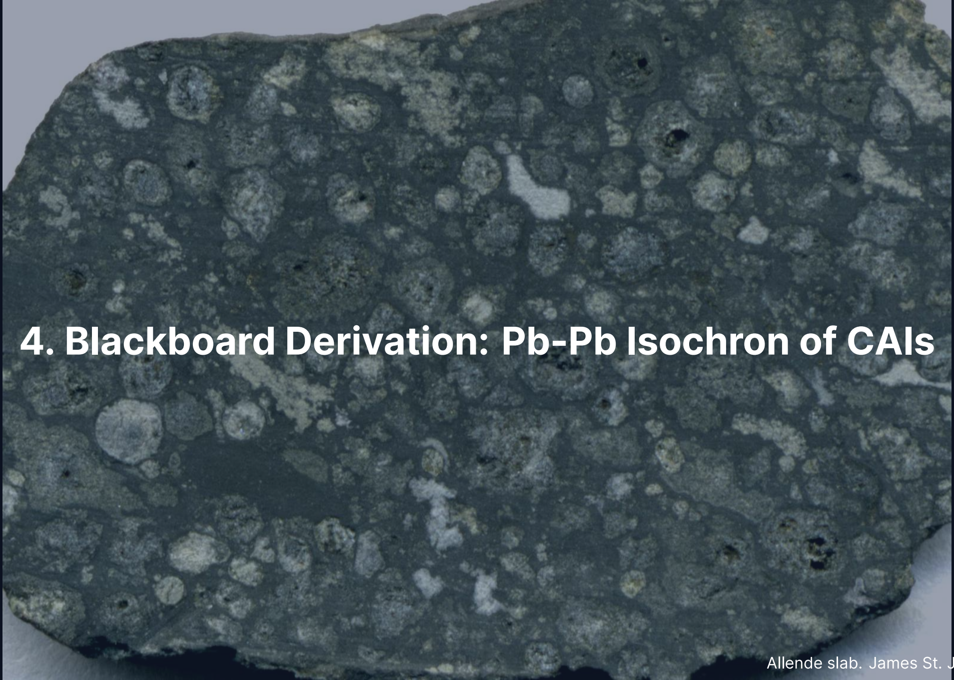
The Early Solar System Timeline

Anchored to the Pb-Pb age of CAIs and cross-calibrated:

- $t = 0$: CAIs condense. 4567.30 ± 0.16 Myr (Connelly et al. 2012).
- $t = 1–4$ Myr: chondrule formation.
- $t = 0.5–4$ Myr: planetesimal differentiation, core formation.
- $t \gtrsim 5$ Myr: large terrestrial protoplanets.

Three independent U/Th-Pb chronometers all give the same CAI age to ±0.16 Myr.

4. Blackboard Derivation: Pb-Pb Isochron of CAIs



Setup: Two Parallel Chains in One Element

- $^{238}\mathrm{U} \to {}^{206}\mathrm{Pb}$ with $\lambda_{238} = \ln 2/(4.4683\,\mathrm{Gyr})$.
- $^{235}\mathrm{U} \to {}^{207}\mathrm{Pb}$ with $\lambda_{235} = \ln 2/(0.7038\,\mathrm{Gyr})$.
- $^{204}\mathrm{Pb}$: stable, no significant radiogenic source. The reference isotope.

Radiogenic daughter at time t since closure:

$$^{206}\mathrm{Pb}^* = {}^{238}\mathrm{U}\,[e^{\lambda_{238}t} - 1], \qquad {}^{207}\mathrm{Pb}^* = {}^{235}\mathrm{U}\,[e^{\lambda_{235}t} - 1]$$

Step 1: Real Rocks Have Initial Pb

Total Pb = initial + radiogenic, so divide by ^{204}Pb:

$$\left(\frac{^{206}\mathrm{Pb}}{^{204}\mathrm{Pb}}\right)_{\text{now}} = \left(\frac{^{206}\mathrm{Pb}}{^{204}\mathrm{Pb}}\right)_0 + \frac{^{238}\mathrm{U}}{^{204}\mathrm{Pb}}\left[e^{\lambda_{238}t} - 1\right]$$

$$\left(\frac{^{207}\mathrm{Pb}}{^{204}\mathrm{Pb}}\right)_{\text{now}} = \left(\frac{^{207}\mathrm{Pb}}{^{204}\mathrm{Pb}}\right)_0 + \frac{^{235}\mathrm{U}}{^{204}\mathrm{Pb}}\left[e^{\lambda_{235}t} - 1\right]$$

Two equations, three unknowns each (t, initial ratio, U/Pb of the rock). *Even taken together, the system is under-constrained.*

Step 2: Use Cogenetic Samples

- Take several mineral fractions of the same rock, all formed at the same time and from the same Pb reservoir.
- All share the same age t and the same initial Pb ratios.
- What *differs* between fractions: the U/Pb ratio (U and Pb partition into different minerals).
- U → apatite, zircon. Pb → feldspar, sulphides. They split.

Step 3: Eliminate U/Pb by Taking the Ratio

Divide the ^{207}Pb equation by the ^{206}Pb equation. The ^{204}Pb terms cancel; the U-isotope ratio simplifies to the present-day $^{235}\mathrm{U}/^{238}\mathrm{U}$:

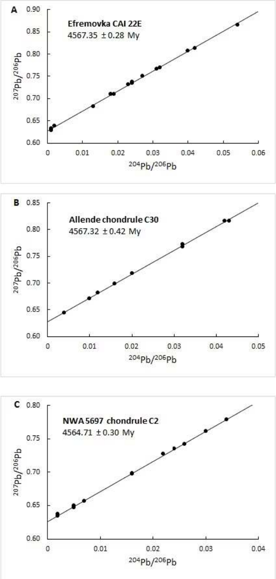
$$\frac{(^{207}\mathrm{Pb}/^{204}\mathrm{Pb})_{\mathrm{now}} - (^{207}\mathrm{Pb}/^{204}\mathrm{Pb})_0}{(^{206}\mathrm{Pb}/^{204}\mathrm{Pb})_{\mathrm{now}} - (^{206}\mathrm{Pb}/^{204}\mathrm{Pb})_0} = \frac{^{235}\mathrm{U}}{^{238}\mathrm{U}} \cdot \frac{e^{\lambda_{235}t} - 1}{e^{\lambda_{238}t} - 1}$$

The right-hand side depends *only on time*. The U/Pb ratio of the rock has dropped out.

Step 4: The Isochron Plot

- Plot ($^{207}\mathrm{Pb}/^{204}\mathrm{Pb}$) vs. ($^{206}\mathrm{Pb}/^{204}\mathrm{Pb}$) for many cogenetic mineral fractions.
- All points lie on a straight line: the **Pb-Pb isochron**.
- Slope → age t (after solving the right-hand-side transcendental equation).
- Intercept → initial Pb composition (drops out automatically; *not* an external assumption).

This is why Pb-Pb is the gold standard: *the double system is self-calibrating*. Single-system U-Pb requires a separate model for initial Pb.



Pb-Pb isochrons: (A) CAI 22E (Efremovka), (B) Allende chondrule C30, (C) NWA 5697 chondrule. From Connelly et al. (2012).

Step 5: Numerical Result

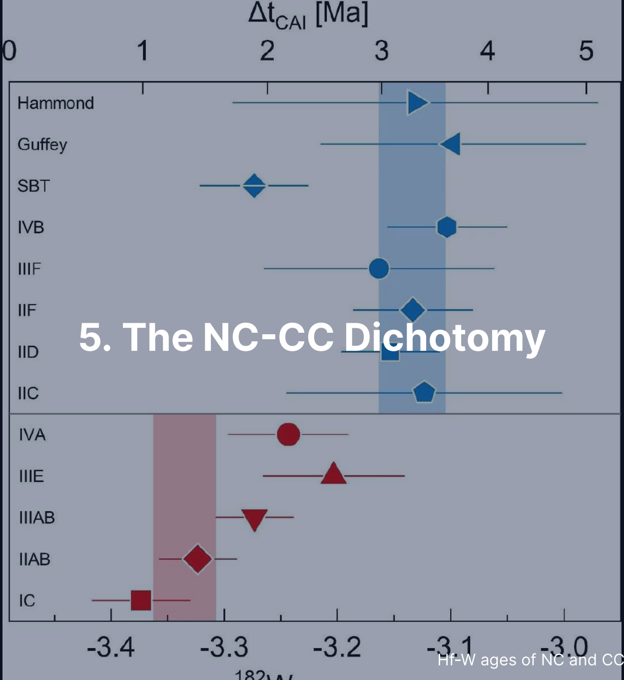
Connelly et al. (2012) measured CAIs from Northwest Africa 2364 (NWA 2364, CV3) after acid leaching:

$$\boxed{t_{\mathrm{CAI}} = 4567.30 \pm 0.16\ \mathrm{Myr}}$$

- Confirmed by independent samples + labs to the same precision (Amelin et al. 2010).
- Chondrules from the same meteorite: ages span a few Myr after CAIs.
- This is the *absolute zero* of the solar system clock.

Two parallel decay chains turned an under-constrained problem into a self-calibrating one.

5. The NC-CC Dichotomy

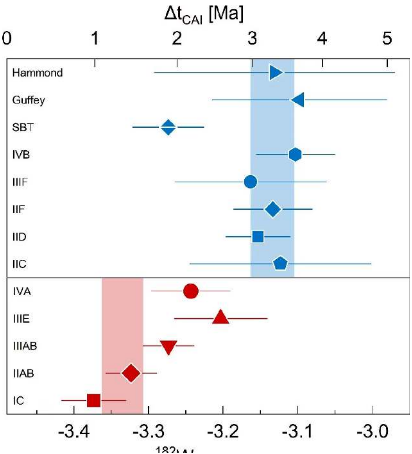


Hf-W ages of NC and CC iron meteorites. Spitzer et al. (2021)

Two Isotopic Reservoirs

- Nucleosynthetic anomalies in $^{50}\mathrm{Ti}$, $^{54}\mathrm{Cr}$, $^{94}\mathrm{Mo}$, $^{100}\mathrm{Ru}$, $^{48}\mathrm{Ca}$ split meteorites *cleanly* into two groups.
- **NC** (non-carbonaceous): ordinary + enstatite chondrites, HEDs, Mars, Earth.
- **CC** (carbonaceous): all carbonaceous chondrites, several iron groups, probably Jupiter and the Trojans.
- Temporally robust: NC and CC of overlapping age are isotopically distinct.
- Did not mix for ~ 3–4 Myr after CAIs.

Hf-W Age Separation



- *Two different x-axes.*
- *Top:* CC irons (blue) vs Δt_{CAI} → cluster at ~ 3 Myr.
- *Bottom:* NC irons (red) vs initial $\varepsilon^{182}\mathrm{W}$ (*not* age) → cluster at ≈ −3.3.
- Hf-W modelling: those NC values → Δt_{CAI} ≈ 0.3–1 Myr.
- NC cores form ~ 2 Myr before CC cores.
- Spitzer et al. (2021).

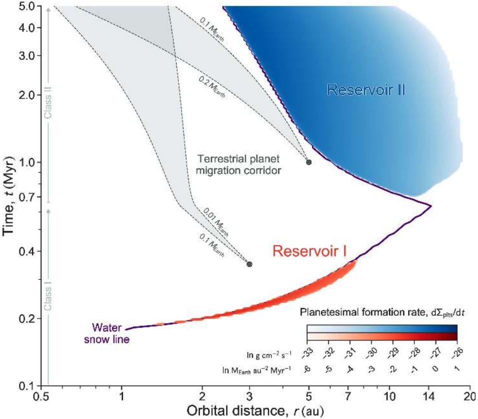
Three Live Interpretations

Same data, three different stories:

1. **Spatial barrier (Kruijer et al. 2017):** Jupiter formed early ($\lesssim$ 1 Myr), opened a gap, stopped CC pebble drift inward. Spatial separation enforces compositional split.
2. **Snow-line + pebble isolation (Lichtenberg et al. 2021):** disk thermal evolution alone makes two epochs of planetesimal formation; Jupiter only required to reach pebble-isolation mass.
3. **Temporal model (Schiller et al. 2018; Nanne et al. 2019; Spitzer et al. 2020):** disk composition itself evolves with time as outer-disk material drifts inward. NC vs CC marks formation *epoch*, not place.

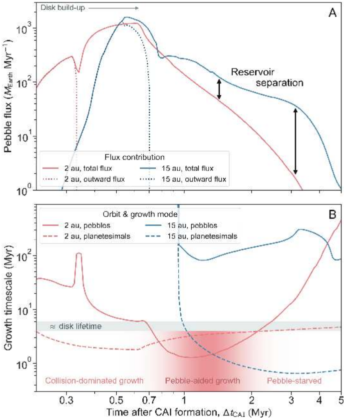
All three match current data. The resolution is one of the major open questions in cosmochemistry.

Snow-Line Mechanism: Two Reservoirs Naturally



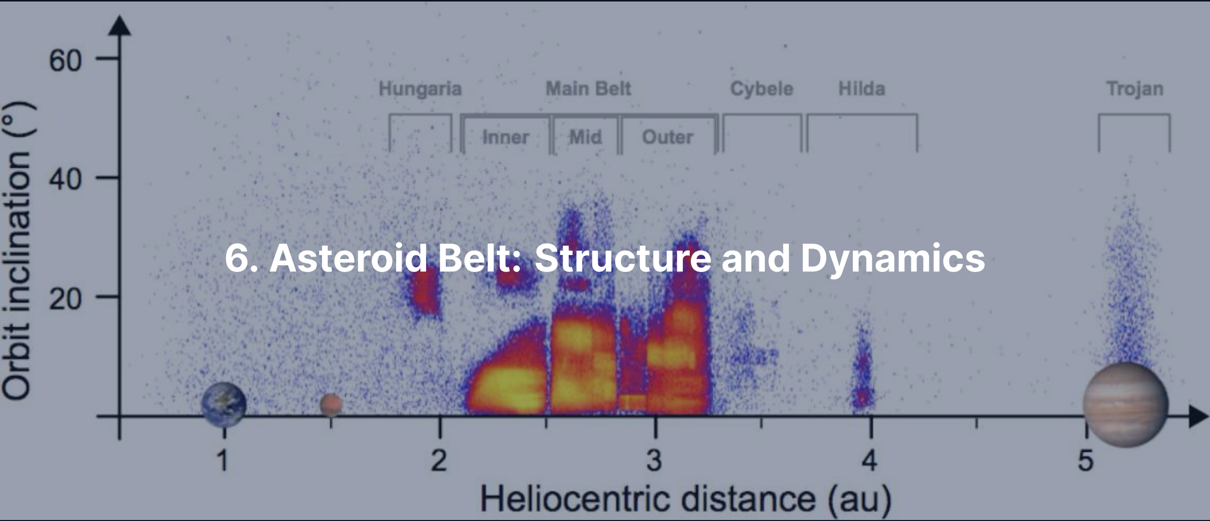
- Disk simulation, time vs. orbital distance.
- Reservoir I (red): early, dry, inner disk = NC.
- Reservoir II (blue): later, ice-rich, outer disk = CC.
- Bifurcation arises from disk evolution itself.
- Lichtenberg et al. (2021).

Pebble Flux Confirms the Two-Reservoir Picture



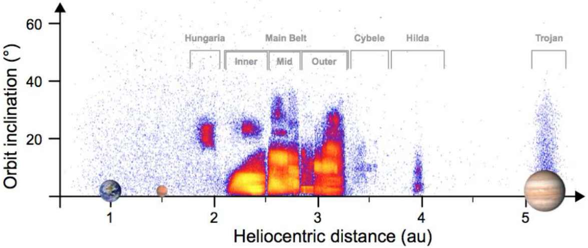
- Top: pebble flux at 2 AU vs 15 AU.
- Reservoirs progressively separate by > 10×.
- Bottom: planetesimal growth timescales.
- ”Pebble-aided growth” interval marked.
- Streaming-instability + pebble-accretion picture.

6. Asteroid Belt: Structure and Dynamics



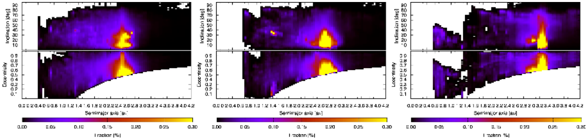
Asteroid orbits in *i* vs *a*. DeMeo & Carry (2014)

The Main Belt in Orbital Space



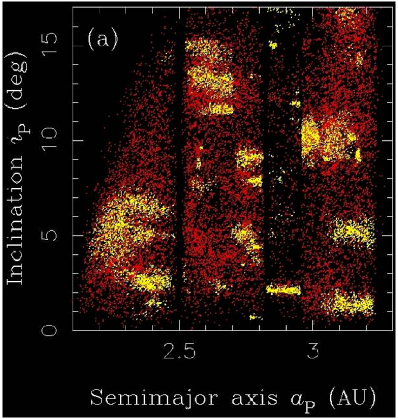
- ~ 1.4 million catalogued bodies; ~ 10^6 at > 1 km.
- Total mass ~ $4 \times 10^{-4}\, M_\oplus$ (Ceres = 30%).
- Hungarias, main belt 2.1–3.3 AU, Hildas, Trojans at 5.2 AU.
- Vertical gaps → Kirkwood resonances with Jupiter.

Kirkwood Gaps as Dynamical Sinks



- Strongest resonances: 3:1 (2.50 AU), 5:2 (2.82), 7:3 (2.96), 2:1 (3.27).
- Periodic Jovian kicks add coherently.
- Eccentricity pumped to planet-crossing in 10^4–10^6 yr.
- Gaps are not empty: they are leaky.
- Granvik et al. (2018).

Asteroid Families = Disrupted Parent Bodies

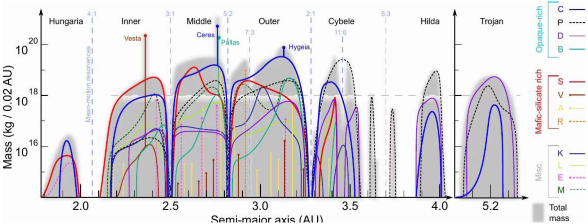


- Hierarchical clustering in proper a, e, i.
- Hirayama families: Themis, Eos, Koronis, Eunomia, Vesta, Flora.
- Each family = debris of a single disruption event.
- Spread → original parent body size, age.

Spectral Classes Track Composition

- **C-type** (carbonaceous, ~ 4% albedo): ~ 75% of outer belt; volatile-rich.
- **S-type** (silicaceous, ~ 15%): ~ 17% of belt; ordinary-chondrite analogues; inner belt.
- **M-type** (metallic): ~ 10%; iron-meteorite or enstatite analogues.
- **V-type**: basaltic 1 μm absorption; the Vesta family.
- **D, P, K**: very dark, very red; transitional outer belt and Trojans.

The Volatile Gradient is Spatially Preserved



- Cumulative mass per 0.02 AU bin.
- S-types: inner belt.
- C, B, P, D-types: outer belt.
- V-type Vesta family at 2.4 AU.
- Maps the disk's snow line.
- DeMeo & Carry (2014).

7. NEAs, Yarkovsky/YORP, and Planetary Defence

Chelyabinsk meteor (2013). A. Ivanov CC BY 3.0

Near-Earth Asteroids

- NEA = perihelion $q < 1.3$ AU.
- ~38,000 known (early 2026); thousands of new discoveries per year.
- PHAs = Potentially Hazardous: $D > 140$ m and Earth-MOID < 0.05 AU. ~2,500 known.
- Typical residence time ~ 10 Myr before ejection or collision.
- Population in steady state → continuous resupply from the main belt.

Three Resupply Channels

- Kirkwood resonances (especially 3:1, 5:2).
- ν_6 secular resonance at the inner belt edge.
- **Yarkovsky drift** into one of the above.

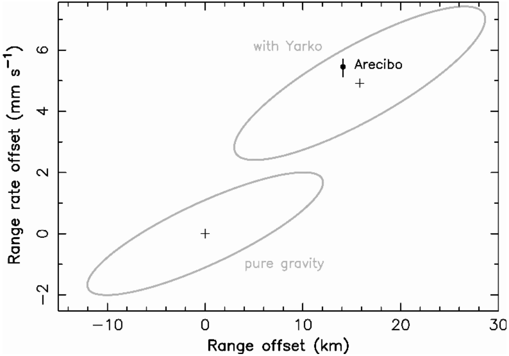
Yarkovsky: a small along-track thermal-recoil force on a rotating body → slow $\mathrm{d}a/\mathrm{d}t$.

Standard scaling for the seasonal Yarkovsky effect:

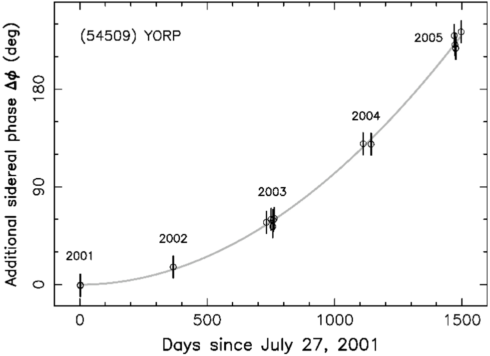
$$\frac{\mathrm{d}a}{\mathrm{d}t} \propto \frac{1}{D\,\rho\,\sqrt{a}}$$

Smaller, less dense, closer-in → drifts faster.

Yarkovsky and YORP Detected on Real NEAs



Yarkovsky: (6489) Golevka, Arecibo radar



YORP: (54509) YORP rotational acceleration

YORP: same thermal asymmetry, acting on *spin*. Can spin small asteroids to fission or stop them.

Impact Frequency, Order of Magnitude

- **10 m** (~ 10 kt): once per 5–10 years, harmless airbursts.
- **20 m** (~ 0.5 Mt): once per decade. Chelyabinsk 2013 was 19 m.
- **50 m** (~ 10 Mt): once per few millennia. Tunguska 1908.
- **140 m** (~ 300 Mt): once per ~ 30 kyr. PHA threshold.
- **1 km** (~ 10^5 Mt): once per ~ 500 kyr. Civilisation-scale.
- **10 km** (~ 10^8 Mt): once per ~ 100 Myr. Mass extinction.

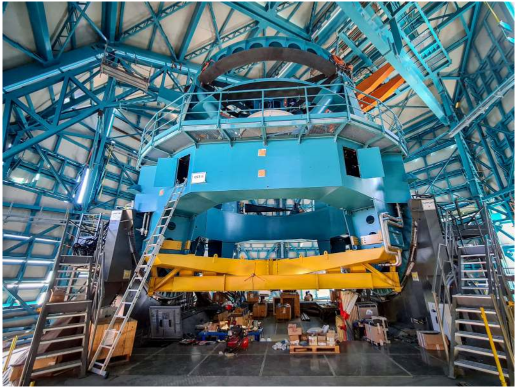
Two Recent Impacts Within Living Memory

Chelyabinsk, 15 Feb 2013

Tunguska, 30 Jun 1908

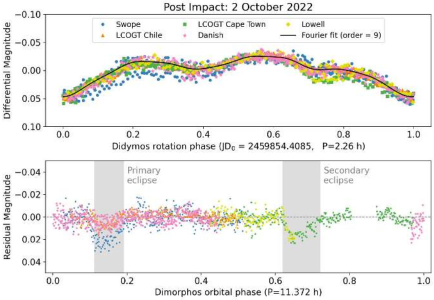
Tunguska levelled ~ 2,000 km^2 of forest. Chelyabinsk shattered windows over a wide area, ~ 1,500 injured. Both airbursts.

Discovery: Vera C. Rubin Observatory



- 8.4 m mirror, 3.2 Gpx camera.
- Engineering first light 2024; First Look 23 Jun 2025.
- Whole sky every few nights for 10 yr.
- Expected: 10× NEA discovery rate.
- Will close PHA inventory > 140 m within a decade.

Deflection: DART, the First Practical Test

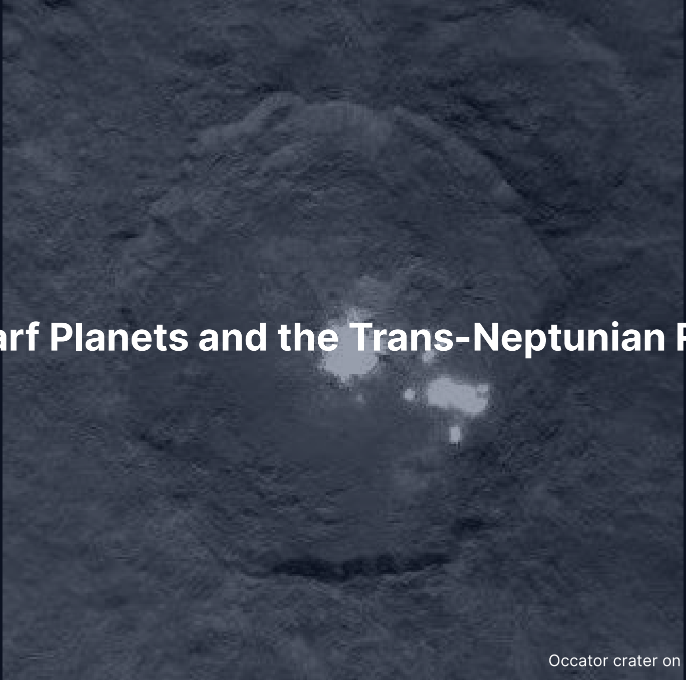


- DART hit Dimorphos (moon of Didymos) on 26 Sep 2022.
- Period reduced by 33.0 ± 1.0 minutes (3σ).
- Momentum-transfer enhancement $\beta \approx 3.6$.
- Ejecta plume recoil > direct impact momentum.
- Hera (ESA, launched 2024): characterise crater + β.

Break

End of Part 1: Meteorites & Inner Small Bodies

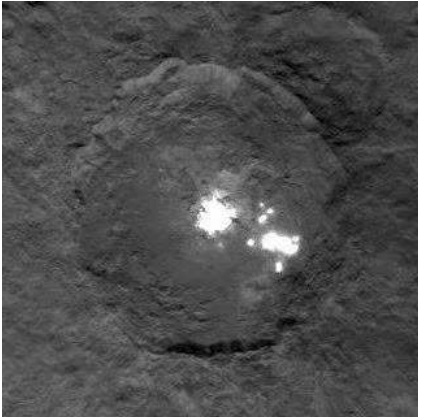
Part 2: Trans-Neptunian Region, Comets, Missions



8. Dwarf Planets and the Trans-Neptunian Region

Occator crater on Ceres, Dawn. NASA/JPL-Caltech

Ceres: Inner-Solar-System Ocean World?

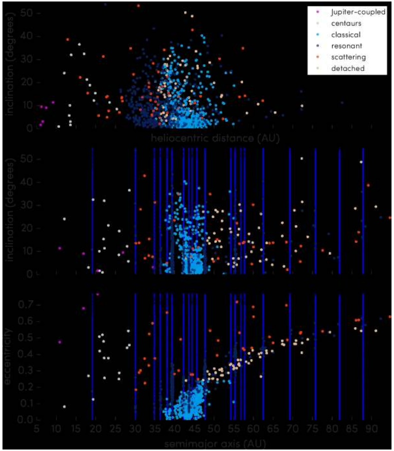


- R = 470 km, mass ~ 1% Moon.
- Dawn (2015–2018): hydrated, ammonia-bearing surface.
- Occator: hydrated $\mathrm{Na_2CO_3}$ + $\mathrm{NH_4Cl}$ from erupted brines.
- Implies (or had) a partially liquid water layer at depth.
- Possible captured outer-solar-system body (Nice-model rearrangement).

Trans-Neptunian Region: Five Sub-Populations

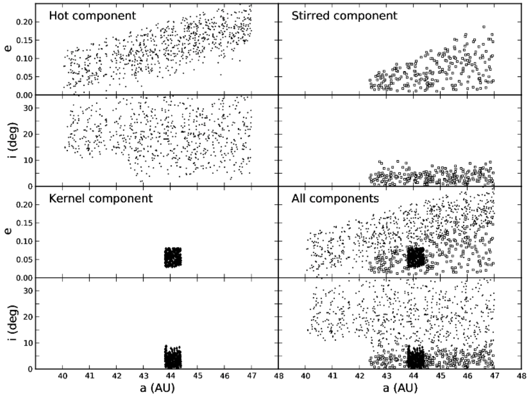
- **Cold classical KB**: $a = 42\text{–}47$ AU, $e \approx 0$, $i \lesssim 5^\circ$. *Primordial.* High binary fraction, very red.
- **Hot classical KB**: same a range, i up to 30°. Scattered-in by Nice-model instability.
- **Resonant** (e.g. plutinos at 3:2, $a \approx 39.4$ AU). Pluto belongs here.
- **Scattered disk**: $q \sim 30$ AU, a to > 100 AU. Still being scattered by Neptune.
- **Detached**: $q > 40$ AU. Decoupled from Neptune. Sedna lives here.

TNO Orbital Architecture (OSSOS)



- 1142 characterised TNOs from OSSOS.
- Vertical lines: Neptune resonances.
- Cold classicals: dense low-*i* cluster at 42–47 AU.
- Bannister et al. (2018).
- Architecture is a fossil of giant-planet migration.

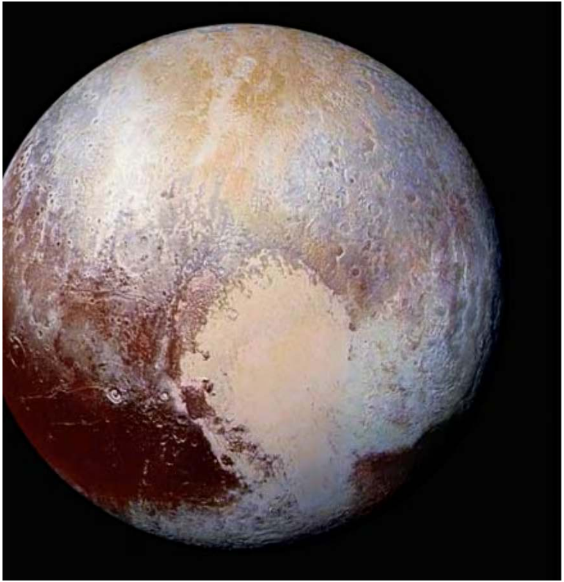
Cold Classicals: Never Disturbed by Neptune



- CFEPS-L7 model, three components.
- Hot, stirred, kernel: distinct (e, i) structure.
- Cold kernel near 44 AU.
- Hot classicals same a but different (e, i).
- → two dynamically distinct origins.

9. Pluto, Charon, Arrokoth

Pluto, New Horizons (2015). NASA/JHUAPL/SwRI

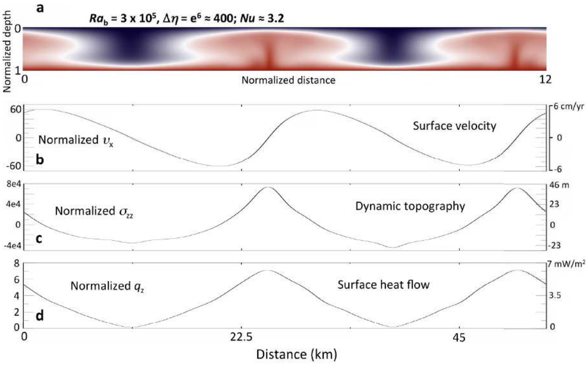


True-colour Pluto, New Horizons Ralph (2015). Tombaugh Regio + Sputnik Planitia.

Reading Pluto

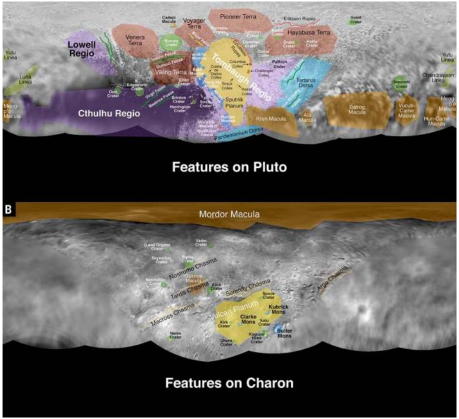
- R = 1188 km, ρ = 1854 kg m^{-3} → ~ 2/3 rock, 1/3 water ice.
- Surface 40 K → water-ice "bedrock", N_2 + CH_4 + CO frosts on top.
- Sputnik Planitia: ~1,000 km N_2-ice basin, *actively convecting*.
- Surface age < 10 Myr.
- Possible subsurface liquid water ocean (Sputnik gravity anomaly + Charon alignment).
- Thin N_2 atmosphere (~ 10 μbar), in slow hydrodynamic escape.

Solid-State Convection in Sputnik Planitia



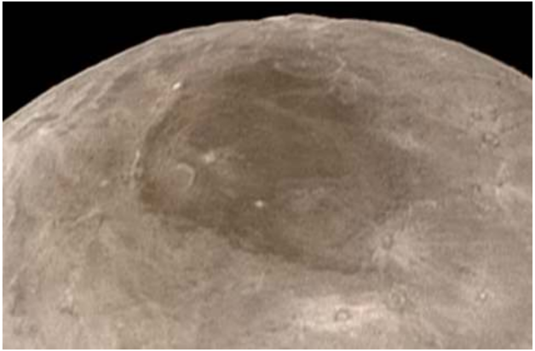
- Numerical model, $\mathrm{Ra}_b \sim 3 \times 10^5$.
- Cells tens of km across.
- Overturn timescale ~ 10^6 yr.
- Driven by present-day radiogenic heat in the rocky core.
- McKinnon et al. (2016).

Pluto-Charon: A Binary World



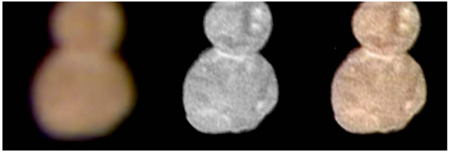
- Charon R = 606 km, ~ 1/8 Pluto's mass.
- Barycentre lies *outside* Pluto.
- Four small moons: Styx, Nix, Kerberos, Hydra.
- Likely fragments of an early collisional event.

Charon’s Mordor Macula



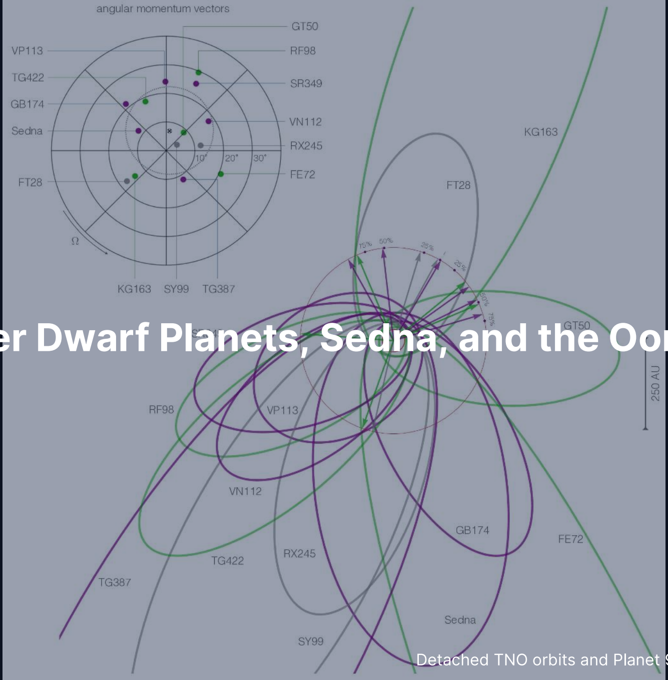
- Reddish-brown polar cap.
- Photochemically processed methane that escaped from Pluto.
- Captured by Charon’s gravity, frozen onto the cold pole.
- Equatorial chasms: extensional tectonics.
- Grundy et al. (2016).

Arrokoth: A Pristine Cold Classical KBO



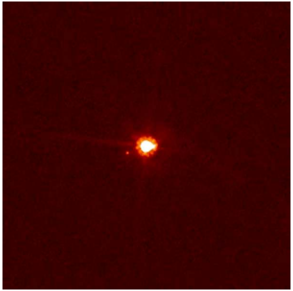
- New Horizons flyby, 1 Jan 2019.
- ~35 km long; **contact binary**.
- Two flat lobes, narrow neck, no impact crater at the join.
- Merger velocity $\lesssim$ few m s^{-1}.
- → formation by streaming-instability gravitational collapse.

10. Other Dwarf Planets, Sedna, and the Oort Cloud

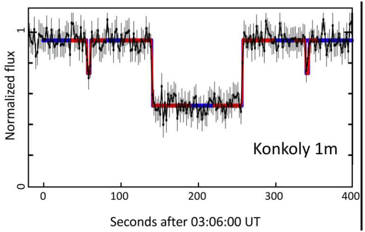


Detached TNO orbits and Planet 9 hypothesis. Batygin et al. (2019)

Eris, Haumea, Makemake



Eris + Dysnomia. More massive than Pluto.

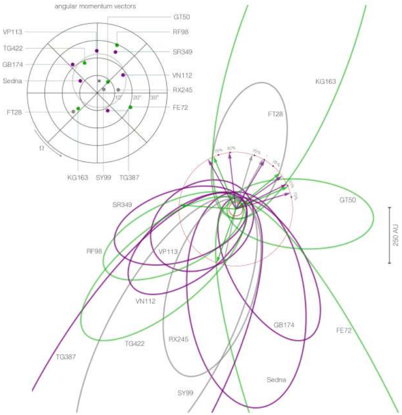


Haumea occultation: revealed a **ring**.

Makemake + small moon. Methane-ice surface.

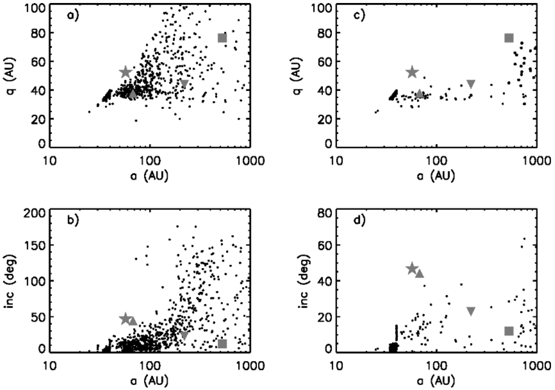
Plus dwarf-planet candidates: Gonggong, Quaoar, Orcus, Salacia, Sedna.

Sedna: Detached, Mysterious



- q = 76 AU, Q = 900 AU. Period 11,400 yr.
- Cannot have been excited by Neptune (q too high).
- Cannot be perturbed by present-day stars (Q too low).
- Possibly inner Oort cloud, or relic of birth-cluster.
- Clustering → **Planet 9** hypothesis (Batygin et al. 2019).

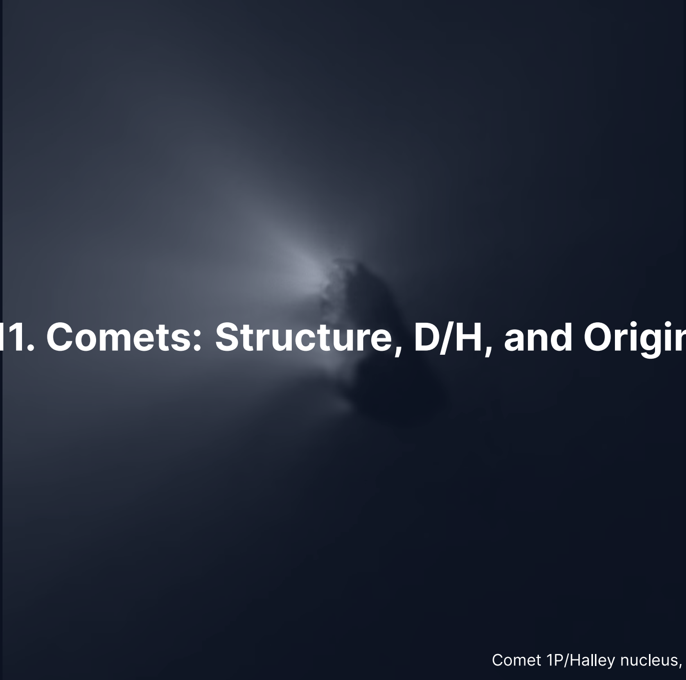
The Oort Cloud



Top row: perihelion q vs semimajor axis a for two simulations of Oort cloud formation. Bottom row: orbital inclination vs a. Kaib & Quinn (2008).

- 2,000–50,000 AU, roughly spherical.
- Inferred 1950 by Oort from long-period comet statistics.
- ~ 10^{11}–10^{12} objects, total mass ~ 1–10 $M_\oplus$.
- Origin: planetesimals scattered out of the Jupiter-Neptune zone, lifted by galactic tide and passing stars.
- **Never directly observed.**

11. Comets: Structure, D/H, and Origin

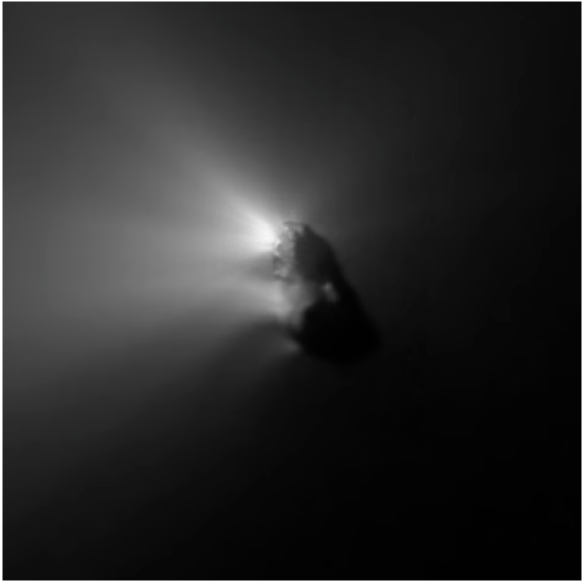


Comet 1P/Halley nucleus, Giotto (1986). ESA/MPAe Lindau

Anatomy of an Active Comet

Whipple's "dirty snowball" (1950), confirmed in detail:

- **Nucleus**: 1–30 km, dark (~ 4% albedo), porous (50–75%), $\rho \sim 0.5$ g cm^{-3}.
- Composition: H_2O (dominant), CO, CO_2, CH_4, NH_3, CH_3OH, HCN, dust, organics.
- Inside ~ 5 AU: water sublimation → activity:
  - **Coma**: 10^4–10^6 km gas+dust envelope.
  - **Ion tail**: blue, straight, anti-sunward (solar wind).
  - **Dust tail**: yellow, curved, lags along the orbit.



Halley nucleus (15 × 8 × 8 km), Giotto Halley Multicolour Camera, 13–14 March 1986. Credit: ESA/MPAe Lindau.

Reading Halley + Halley from Earth

Halley from ESO, March 1986. Long tail, bright coma.

- Period 76 years.
- Nucleus 15 × 8 × 8 km, irregular, very dark.
- Active jets confined to small surface fractions.
- Halley-type comet ($T_J \approx -0.6$, retrograde): dynamically Oort-cloud-derived but captured to a 76-yr orbit.

Short-Period vs Long-Period: The Tisserand Parameter

$$T_J = \frac{a_J}{a} + 2\sqrt{\frac{a}{a_J}(1 - e^2)}\cos i, \qquad a_J = 5.20\,\mathrm{AU}$$

- Conserved across close encounters with Jupiter (Jacobi integral).
- $T_J > 3$: main-belt asteroids decoupled from Jupiter.
- $2 < T_J < 3$: Jupiter-family comets.
- $T_J < 2$: long-period (Oort cloud) comets.

Examples: Ceres $T_J \approx 3.3$, 67P $T_J \approx 2.7$ (JFC), Halley $T_J \approx -0.6$ (LP).

D/H of Cometary Water vs Earth

103P/Hartley 2 (EPOXI 2010): D/H ≈ VSMOW.

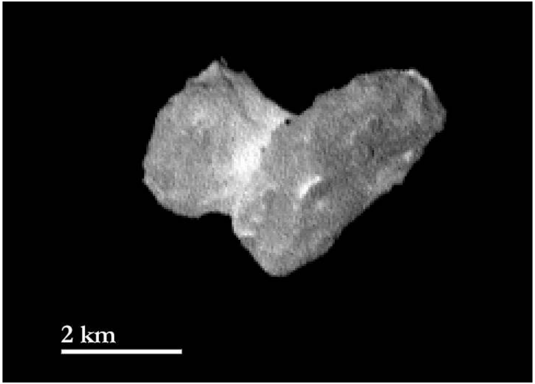
- Earth oceans: D/H = 1.56×10^{-4} (VSMOW).
- Hartley 2 (JFC): ≈ VSMOW.
- 67P (JFC, Rosetta): ≈ 3× VSMOW.
- Long-period comets: ≈ 2× VSMOW.
- Comets span 3× in D/H. **No clean class match for Earth.**

Conclusion: Earth's water ≠ "delivered by comets". Carbonaceous chondrites must dominate; comets are a small tail.

12. Recent Missions and Sample Return

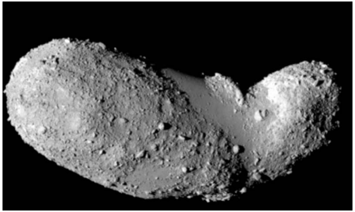
Comet 67P/Churyumov-Gerasimenko, Rosetta. ESA/Rosetta/NavCam

Rosetta + Philae at 67P (2014–2016)

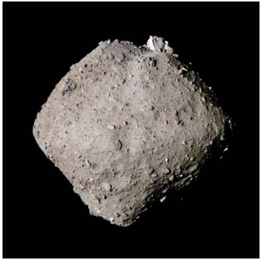


- First spacecraft to orbit a comet, first lander on one.
- 67P bilobed ”rubber duck”: contact binary, gentle merger.
- Bulk density 0.533 g cm^{-3}; porosity ~ 70–80%.
- D/H = 5.3×10^{-4} (~ 3× VSMOW).
- Detection of glycine + phosphorus + prebiotic molecules.

Three Sample Returns: Itokawa, Ryugu, Bennu



Hayabusa, 2010: 1,500 grains. Itokawa = LL chondrite parent.

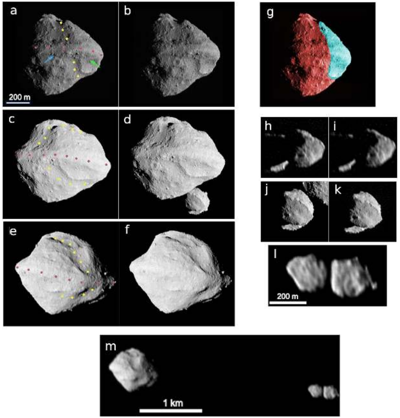


Hayabusa2, 2020: 5.4 g. Ryugu ≈ CI chondrite. Amino acids, nucleobases.

OSIRIS-REx, 2023: 121 g. Bennu CM/CI-like, Mg carbonate veins.

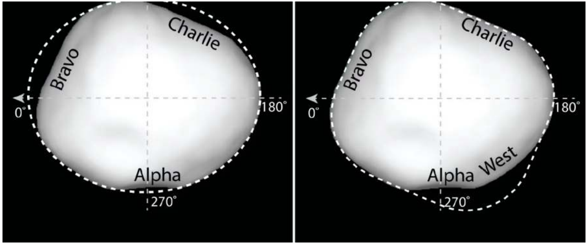
Each rock → specific parent body with a known orbit and full in-situ context.

Lucy: Tour of the Trojans



- Launched Oct 2021. 12-yr tour.
- Eight Trojans across L4 + L5 swarms (2027–2033).
- D-type Trojans: probably Nice-model captured KBOs.
- Surprise at Dinkinesh (Nov 2023): satellite Selam = contact-binary moonlet.

Psyche: A Metal World



- Largest M-type asteroid, ~ 226 km.
- Launched Oct 2023. Arrival 2029.
- Hypothesis: **exposed core** of a destroyed differentiated planetesimal.
- Test: remnant magnetisation → once-active dynamo.
- Plus first deep-space optical-comm experiment.

13. Interstellar Visitors

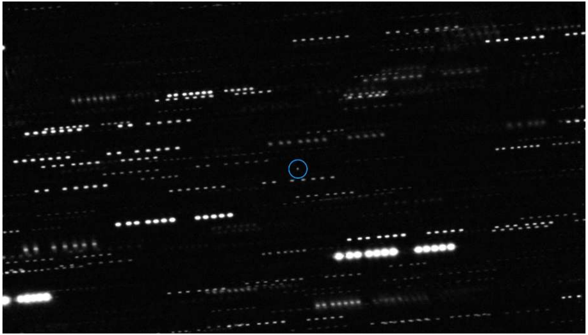
1I/'Oumuamua deep image. ESO / K. Meech et al.

Three ISOs in Eight Years

- **1I/'Oumuamua** (Oct 2017, Pan-STARRS): ~100–200 m, elongated, no detectable coma, non-gravitational acceleration on exit.
- **2I/Borisov** (Aug 2019): unambiguously cometary. Coma + dust tail. CO/H_2O *much higher* than solar-system comets.
- **3I/ATLAS** (Jul 2025): larger, clearly active. Initial characterisation ongoing.

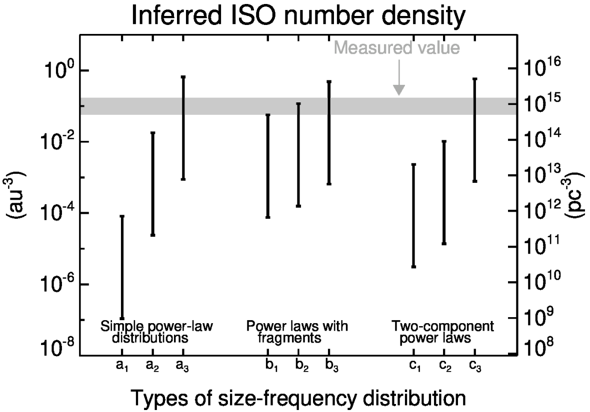
Pre-discovery population estimates were wildly low.

'Oumuamua: First and Strangest



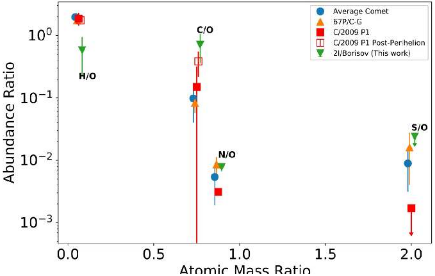
- Centre: 'Oumuamua. Stars trail because the telescope tracked the target.
- No detectable coma at the limit of stacked imaging.
- Yet outgoing trajectory shows non-gravitational acceleration.
- Nature still debated: H_2 ice fragment? N_2 ice? Tidal disruption shard?

Implied ISO Population is Large



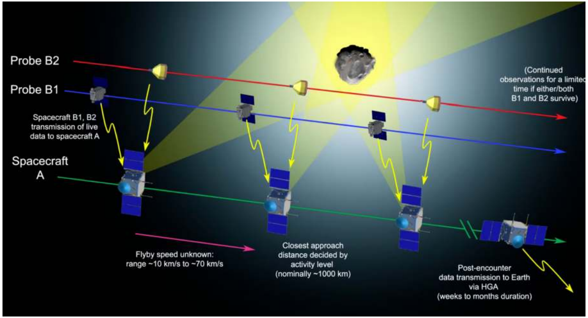
- Detection of one ISO with Pan-STARRS implies ~ 10^4 ISOs > 100 m within Neptune's orbit *at any moment*.
- Each ISO is a sample from a different planetary system.
- Cumulative ejection rate: each star ejects planetesimals at rates similar to ours.

Borisov: First Interstellar *Comet*



- Active comet from outside the solar system.
- HST/COS coma composition (Bodewits et al. 2020).
- Decisively CO-enriched relative to H_2O.
- → formed in colder region than solar-system comets.

ESA Comet Interceptor: A Standby Mission



- Planned launch 2029.
- Parked at Sun-Earth L2.
- Waits until a suitable target is discovered (long-period or interstellar).
- Three flyby probes for 3D reconstruction.
- Turns transient ISO discoveries into intercepts.

Summary: Today's Course-Level Takeaways

1. Small bodies are **formation fossils**: composition + dynamics still record the solar system's earliest history.
2. Pb-Pb dates the oldest solids at 4567.30 ± 0.16 Myr. Two parallel U decay chains make the method self-calibrating.
3. NC-CC dichotomy: same data, three live interpretations (spatial / dynamical / temporal). One of cosmochemistry's biggest open problems.
4. Asteroid belt + KB orbital architecture is a fossil of giant-planet migration.
5. Saturn's Roche limit, Jupiter's Hill sphere, the Tisserand parameter: same conservation-law toolkit applied across populations.
6. Sample return + DART + Lucy + Psyche + Rubin + Comet Interceptor: cosmochemistry is becoming a comparative science across stars.

Questions?

Next: **Lecture 13. Exoplanets**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`